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ORIGINAL PAPER

Root anchorage of hinoki *(Chamaecyparis obtuse* (Sieb. Et Zucc.) Endl.) under the combined loading of wind and rapidly supplied water on soil: analyses based on tree-pulling experiments

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Abstract Wind damage in Japan is mainly caused by typhoons (i.e., tropical cyclones), which are characterized by intensive heavy rainfall and strong winds. In this study, we conducted tree-pulling experiments on two sites to find out whether rapidly supplied water on the soil would affect stability of root anchorage of hinoki (Chamaecyparis obtuse (Sieb. Et Zucc.) Endl.), as expected. For the experiments, we first supplied several quantities of water around the target trees, and then they were pulled down using a wire winch. On study site 1 (Kamiatago experimental forest), we applied general tree-pulling experiments (no water supply) in 2008 and six different irrigation treatments around the target trees in 2009. On study site 2 (Chiyoda experimental forest), we applied one irrigation treatment in 2009. As a result, five trees were uprooted and two were broken in 2008, and all nine trees were uprooted in 2009 on study site 1, regardless of irrigation treatment. On study site 2, two trees pulled down after 4 h of water supply were ruptured at the stem base, opposite to two trees pulled down immediately after supplying water. The water content below the root plate significantly affected root anchorage and more specifically, the maximum turning moment, stem angle at the maximum force, and stiffness index. Moreover, water inside the root plate increased root anchorage at the beginning of a tree failure process. However, it also reduced the root plate area of the hinge side. Conversely, high water content below the root plate decreased root anchorage.

Keywords Tree-pulling experiments · Wind damage · Root anchorage · Water content in the soil

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Introduction

Typhoons, the local name for tropical cyclone in the western Pacific, cause catastrophic damage in forested areas due to their strong wind and heavy rain. In addition, typhoons occasionally enhance the activities of autumn rain fronts, which are conflicts between warm air from southern regions and cold air from northern regions. The combined effects of a typhoon and a rain front cause tremendous rainfall and strong winds. For example, during Typhoon No. 18 in 1991, which caused catastrophic damage in forests, daily precipitation was approximately 250 mm in the Fukuoka Prefecture located in the Kyusyu Island, Japan. In the Owari region, Mie Prefecture, the total precipitation was 545 mm from 17 to 20 September due to the typhoon and the rain front (Ogura 1994). Such meteorological phenomena lead to enormous economical and environmental losses. Approximately 22 million m^3 of timber volume in 1954, 27 million m^3 in 1959, and 11

million m^3 in 1991 were damaged mainly due to catastrophic typhoons (Suzuki et al. 2009).

Kuboyama et al. (2003) pointed out that wind damage will increase in Japanese forests, because a lot of planted forests will reach their maturity in the near future (i.e., age of trees with highest vulnerability to wind pressure). Moreover, typhoons may become larger under the climate change and may also be more destructive (Meehl and Stocker 2007). Previously, field investigations have been carried out after catastrophic wind damage in order to understand which factors contributed to damage. For example, Noda (1993) studied, in forests damaged by a typhoon in 1991, the occurrence of different failure types, and effects of species and terrain conditions on damages. The relationship between wind direction and tree damage has also been analyzed based on forest damage data (Shichiri 1987). However, the mechanisms of tree failure due to typhoon are still poorly understood because of the limited amount and type of information regarding damage (Kamimura and Shiraishi 2007).

In addition to field investigations in damaged areas, tree-pulling experiments have also been conducted to increase the understanding of the mechanism of tree failure due to strong wind. For this purpose, a tree is mechanically pulled down using a wire rope and a winch to demonstrate an applied force of wind on the stem. This provides information on root anchorage and other tree factors associated with wind damage (Nicoll et al. 2006). In Japan, several commercial coniferous tree species have been analyzed using tree-pulling experiments (e.g., Kamimura et al. 2007; Kato 2001; Koizumi 1987; Morioka and Kitagawa 1987; Nonoda et al. 1996). These studies have mainly focused on the mechanism of tree failure due to wind or snow for particular tree species and environmental situations.

Tree-pulling experiments also reveal how the stability of root anchorage is affected by stand, terrain, and soil conditions. For example, Cucchi et al. (2004) found that Maritime pine (*Pinus pinaster*Ait.) at the forest edge had 20% more resistance to uprooting than those situated inside the forest. Nicoll (2005) has also argued that the turning moment of a tree pulled down the slope is significantly lower than that observed for the upslope. In addition, trees are more likely to be broken than uprooted during frozen soil than unfrozen soil and vice versa, and higher turning moments are required to damage trees during the frozen soil (Peltola et al. 2000). These studies indicate that vulnerability of root anchorage is depending both on species characteristics and on environmental circumstances. Such research findings have led to the development of mechanistic models for wind damage risk assessment such as GALES and HWIND (Gardiner et al. 2008, 2000; Peltola et al. 1999). GALES could also be used for Japanese forests with reference to supporting decision-making for sustainable forest management (Kamimura et al. 2008).

However, previous experimental work and model development do not account for the rapid change in meteorological conditions (in particular, rainfall before strong winds during a typhoon) that might affect root anchorage due to high water content in the soil. If the aggregate composition is unstable, then rapid wetting of the soil is more likely to accelerate soil breakage. When pores near the soil surface are suddenly filled with water, air is trapped in the deeper soil. This air, which is under pressure by the water near the surface, can break the aggregate composition (Ashman and Puri 2002). When Koizumi (1987) previously conducted tree-pulling experiments after rainfall (1 day after 68 mm of precipitation) on Hokkaido Island, he found lower tree stability than on the corresponding soil not receiving rainfall. Nevertheless, it is not clear how soil water content affects tree stability and which factors relate to root anchorage in the soils associated with rapid rainfall in addition to strong wind pressure.

When a tree receives pressure from wind, soil resistance is the most significant component at the beginning of failure. Subsequently, root anchorage is associated with windward roots, root plate weight, and hinge (Coutts 1986). We assume that as a result of heavy rainfall and as soil rapidly getting much water. These components might be affected by additional factors such as the weight of water and soil resistance to penetration. In this work, we conducted tree-pulling experiments on two study sites in Japan to find out whether rapidly supplied water (applying different irrigation treatments) on the soil would affect stability of root anchorage of hinoki (*Chamaecyparis obtuse* (Sieb. Et Zucc.) Endl.). Such analyses on root anchorage can help to understand in more detail about complex mechanisms of tree failure related to typhoons associated both with strong winds and with heavy rainfall, and to further develop mechanistic wind damage risk assessment models in this respect, too.

Materials and methods

Site description and selection of experimental trees

As shown in Fig. 1, there were two study sites used for the tree-pulling experiments in hinoki (*Chamaecyparis obtuse* (Sieb. Et Zucc.) Endl.), which is a valuable commercial species that is often planted for timber supply in Japan. Table 1 shows the stand, soil, and experimental details.

Study site 1 is located in the Kamiatago experimental forests owned by Shizuoka University, Shizuoka, Japan (34°54′27″ N, 137°44′ 41″ E). The main soil type is referred to as a yellowish brown forest soil. Elevation is 340 m above sea level (a.s.l.); slope is 20°; and slope aspect is east-southeast. There were two tree-pulling

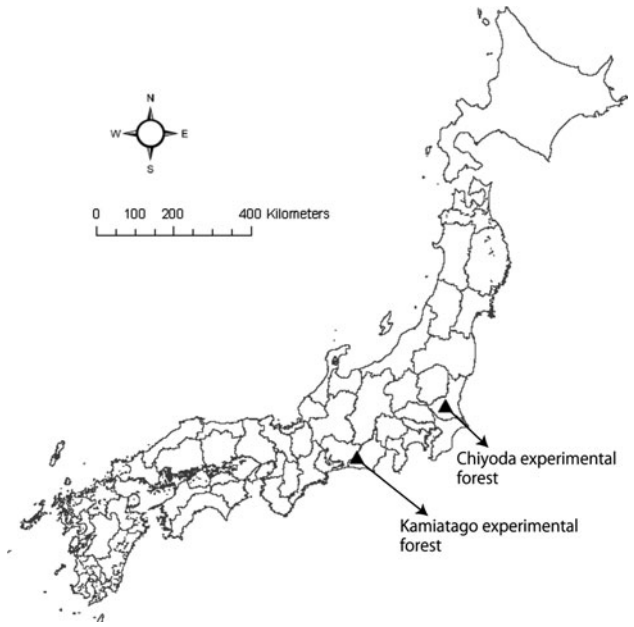


Fig. 1 Location of study site 1 (Kamiatago experimental forest, Japan) and study site 2 (Chiyoda experimental forest, Japan)

Table 1 Description of study layout for the tree-pulling experiments

Location	Study site 1		Study site 2
	Kamiatago, Shizuoka Prefecture		Chiyoda, Ibaraki Prefecture
Experiment year	2008	2009	
Experiment date	12–13 Jun. and 12 Sep.	22 May–4 Jun.	17–18 Dec.
Number of sample tree	7	10	6
Tree age	30	30	12
Mean tree height (m)	15.8	16.4	6.7
Mean *dbh* (cm)	19.5	18.1	12.7
[Treatments]			
Provided water (mm)	0	12.5, 25, 50, 100, 200, 400	100
Number of slope sensor	3	3	2

experiments conducted at study site 1. The first experiment was carried out in 2008, without supplying water. At the time of these tree pullings, mean temperature was about 18.6°C on 12 through 13 June and 22.5°C on 12 September 2008 (with a precipitation of 7.5 mm in latter case). No other rainfall was recorded 2 days before and during these experiments. Seven hinoki trees (about 30 years old) were pulled down at the first and second rows from the forest edge for the tree-pulling experiments.

The second tree-pulling experiment was carried out on study site 1 on 1 through 4 June 2009. Mean temperature at the time was approximately 17°C. No major rainfall occurred prior to the tree-pulling experiments. During the experiments, there was rainfall of 3 mm during the night of 3 June 2009. Ten hinoki trees (about 30 years old) were pulled down in this experiment.

Study site 2 is located in the Chiyoda experimental forest managed by the Forestry and Forest Products Research Institute, Ibaraki, Japan (36°11′2″ N, 140°13′ 20″ E). The main soil type is an Andisol soil (Nagakura et al. 2006). Elevation is 41 m a.s.l. and the terrain is flat. Tree-pulling experiments were conducted on 17 through 18 December 2009. There was no rainfall in this study area on 12 through 18 December. Strong wind from any particular direction was not recorded. Six 12-year-old hinoki trees were selected for the study. They were genetically identical and planted at the same time in a single row; the distance between neighboring trees was 3 m. Because the root growth was not affected by the terrain slope and genetic bias of the trees could be ignored, impacts from the experimental settings were minimized compared with study site 1.

Tree-pulling experiments

The procedure for the tree-pulling experiments was largely based on previous studies (e.g. Nicoll et al. 2005, 2006). A tree was mechanically pulled down using a wire cable of 10 mm in diameter. Two or three inclinometers (D5R-L02-60, OMRON Co., Tokyo, Japan) were set on the stem at the attached wire height and stem base for the trees on both study sites and at the half of the wire height for those in Study site 1. A five-ton load cell (LT-5TSA71C, Kyowa Electronic Instruments Co., Ltd., Tokyo, Japan) was inserted between the end of the wire cable and the winch of a logging machine for study site 1 and a 3-t manual winch for study site 2. The slope sensors and the load cell were connected to a data logger (GL200A, Graphtec Co., Kanagawa, Japan) in order to record the data at the same time.

Different arrangements were used for the tree-pulling experiments in the respective study sites because tree size and terrain conditions differed. On study site 1, a wire was attached at approximately half of the tree height. A pulley block was also set at the same height, and the wire cable was attached to an anchor tree with another pulley block in order to reduce the required pulling force. Because differences in root anchorages were expected if artificially pulling down or up a tree on slope conditions (see e.g., Nicoll et al. 2005), trees were pulled down parallel to contour lines. For study site 2, no regulation of pulling direction was used because of flat terrain. The wire attachment height was set to under the lowest live whorl (approximately 2 m in height). The wire was directly pulled using a manual winch. In addition, root width and

depth as well as tree weight were measured after pulling down the target tree.

Tree-pulling experiments in 2009 were conducted with supplying additional water to the soil. There were two kinds of approaches to supply water on the soil around the target trees. For study site 1, a 2 × 2 m wood frame covered with a plastic sheet was set around a target tree in order to prevent water from running off, because the site was located on a slope. A 1,000-l water tank was set near the target trees, and water was supplied using a hosepipe from the tank to the inside of the wood frame. Considering the precipitation that would result from a typhoon, the quantities of water supplied to the soil were 12.5, 25, 50, 200, and 400 mm (see Table 1). Because over 400 mm of rainfall was recorded in Aichi Prefecture in 2000 as a result of a typhoon and a rain front (Tanaka 2000), up to 400 mm of water was supplied in this study. The trees were pulled down immediately after supplying water on study site 1.

On study site 2, a 1.5 × 1.5 m wood frame was tentatively set around a target tree to identify the area for supplying water. A 250-l water tank was placed near the target trees. Water was supplied to four trees. Two trees did not receive water, so that they could be used for comparison. Two trees with supplied water were pulled down immediately after providing the water; the other two trees were pulled down 4 h after supplying the water in order to ensure that the soil was saturated with water.

Before pulling trees over, tree characteristics such as tree height and diameter at breast height (dbh) were measured first. Then, the surface soil moisture was determined using a hydro sensor (HydroSense TM, Campbell Scientific Australia Pty. Ltd.) at four points inside the frame at 20 cm depth from the soil surface before and after supplying water. Soil moisture was further measured below (20 cm depth from the bottom of the root plate) and inside (20 cm inside from the bottom of the root plate) the root-soil plate after pulling down the tree. Next, soil resistance to vertical penetration was measured using a soil penetrometer (DIK-5521, Daiki Rika Kogyo Co., Saitama, Japan) at two points inside the frame from the soil surface down to 50–80 cm depth before and after supplying the water.

Finally, plastic flags, approximately 20 cm in height, were placed 10–20 cm apart after supplying water to monitor the soil movement on the root plate. Two video cameras and a digital camera were set around the target tree to record flag movement (Fig. 2). Later, the flag positions were plotted to create grid data, including the area of soil movement at the maximum force, root plate area, and area of hinge (leeward) side. These calculations were conducted using GIS software, IDRISI Andes (Clark Labs., Worcester, USA)

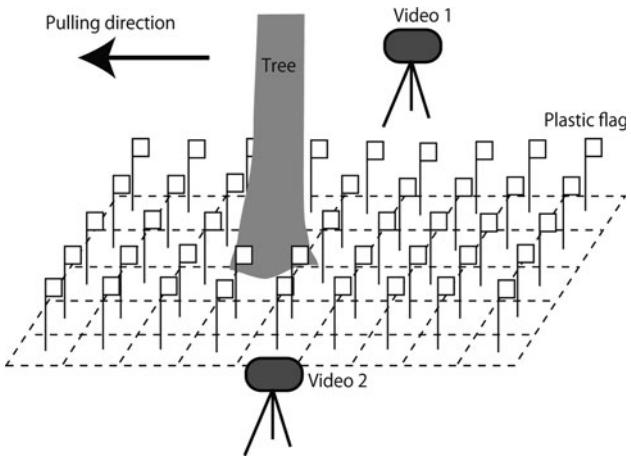


Fig. 2 Description of measuring root plate movement

Data analysis

The maximum turning moment at the stem base (TM_{max}, Nm) consists of the horizontal (applied) forces ($TM_{applied}$, Nm) by pulling and the vertical force ($TM_{gravity}$, Nm) caused by the self-weight of a tree, which were calculated as follows:

$$TM_{applied} = FH_{cable}(\cos\alpha \sin\beta + \sin\alpha \cos\beta) \quad (1)$$

where F (kgf) is the maximum load, α (degree) is angle between the wire and the horizontal line from the stem base of the anchor tree. β (degree) is mean stem angle from the original position of stem at the maximum force, and H_{cable} (m) is the attached wire height.

$$TM_{gravity} = WG \quad (2)$$

where W (kg) is the tree weight, and G (m) is the horizontal displacement of the center of gravity of a tree.

In addition, the areas of soil movement at the maximum force and of hinge side were normalized against the area of root plate. Root plate volume was determined using mean plate depth and plate diameter under the assumption of root plate as a cylinder. Stiffness index (SI) was calculated as follows (see also Koizumi 1987):

$$SI = TM_{max}/\beta \quad (3)$$

The relationship between the root anchorage and soil conditions due to water supply on the soil was studied by applying Pearson's correlation coefficient (r). Characteristics relating to root anchorage such as TM_{max}, $TM_{applied}$, $TM_{gravity}$, stem angle from the original position (β), and SI were explained using tree characteristics such as dbh, tree height, multiplied dbh × tree height (dbh^2h), and tree stem weight (fresh). Root plate characteristics (area and volume) and soil water condition (water contents in the soil, inside, and below the root plate) were also measured.

To normalize the soil movement area, the root-soil area on the hinge side and soil movement area at the maximum force were divided by the total root plate area.

The data were statistically analyzed using SPSS 16.0 and 18.0 (SPSS Inc., Chicago, USA). For the soil compression strength, a paired t-test was conducted to indicate whether the strength changed before and after supplying the water. Analysis of covariance (ANCOVA) was also performed to investigate differences in tree stability (*SI* against dbh^2h) in study site 1 with and without additional water supply (data collected in 2008 and 2009).

Results

The main results of the tree-pulling experiments are shown in Table 2. Due to technical errors, one tree in the 2009 experiments at study site 1 and one tree at study site 2 could not be used for further data analysis. Two of the target trees were broken in 2008 (without supplying water). In 2009, all trees were uprooted in study site 1, whereas three of five trees were broken in study site 2. One of the broken trees had no water supply in the soil, and the others were supplied water 4 h before tree pulling. These results indicate that all trees receiving additional water and pulled down immediately after water supply were uprooted. Moreover, β was not different among uprooted trees although there were various amount of soil water and tree characteristics used for the experiments. In addition, stem angle of the uprooted trees was smaller for the broken trees. Normalized soil movement area at the maximum force of the broken trees was also much larger than that observed for the uprooted trees, although the normalized hinge-side area did not seem to be different between the failure types. *T*rees receiving water on the soil resulted in reduced tree stiffness (see Fig. 3, $P < 0.05$).

As shown in Fig. 4, compressive strengths were plotted based on two depth groups; soil surface (10–30 cm) and deeper soil (>30 cm). A depth of 30 cm was used for grouping because of the difference in wood biomass density. Over 75% of underground biomass is present in the soil surface down to 30 cm depth (Montague and Day 1980). In the surface soil (10–30 cm depth) of study site 1, the mean compressive strength was significantly ($P < 0.05$) higher before water supply (11.0 ± 3.6 kgf/cm^2) than after it (8.8 ± 2.9 kgf/cm^2). Also in the deeper soil (>30 cm), the mean compressive strength was significantly ($P < 0.01$) higher before water supply (13.5 ± 2.3 kgf/cm^2) than after it (12.1 ± 2.2 kgf/cm^2). In the surface soil of study site 2, the mean compressive strength was also significantly ($P < 0.01$) higher before water supply (17.7 ± 3.1 kgf/cm^2) than after it (11.1 ± 4.2 kgf/cm^2). In the deeper soil, the difference in mean compressive strength before and

Table 2 Average values and the standard deviation (SD) from the tree-pulling experiments

Location	Experiment year	Water supply[a]	Failure	*n*	*dbh* (cm)	Tree height (m)	Stem angle (°)	TM_{max} (Nm)	Stem stiffness index	Normalized soil movement area[b]	Normalized hinge-side area[c]
Study site 1	2008	No	Uproot	5	18.7 (*4.0*)	15.4 (*1.9*)	20.3 (*8.0*)	42215.4 (*21577.0*)	2373.5 (*1374.6*)	–	–
			Broken	2	21.6 (*0.6*)	16.8 (*0.8*)	24.2 (*8.8*)	72258.6 (*18998.1*)	3047.1 (*325.4*)	–	–
Study site 1	2009	Yes	Uproot	9	18.3 (*2.0*)	16.6 (*0.6*)	23.3 (*8.0*)	30825.4 (*5812.0*)	1501.3 (*634.8*)	213.2 (*150.9*)	14.9 (*10.1*)
Study site 2	2009	Yes	Uproot	2	11.5 (*0.7*)	6.7 (*0.3*)	22.3 (*9.0*)	9252.2 (*977.9*)	460.4 (*228.4*)	70.7 (*13.1*)	30.0 (*0*)
			Broken	3	13.0 (*2.6*)	6.4 (*1.1*)	18.5 (*10.0*)	11941.0 (*5448.0*)	826.4 (*10.0*)	516.7 (*306.4*)[d]	33.3 (–)[e]

No data of soil movement and hinge-side area were collected in 2008

[a] Supplying water in the soil before pulling down a tree

[b] Normalized soil movement area at the maximum force against the root plate

[c] Normalized hinge (leeward) side area against the root plate

[d] The number of data is 2

[e] The number of data is 1

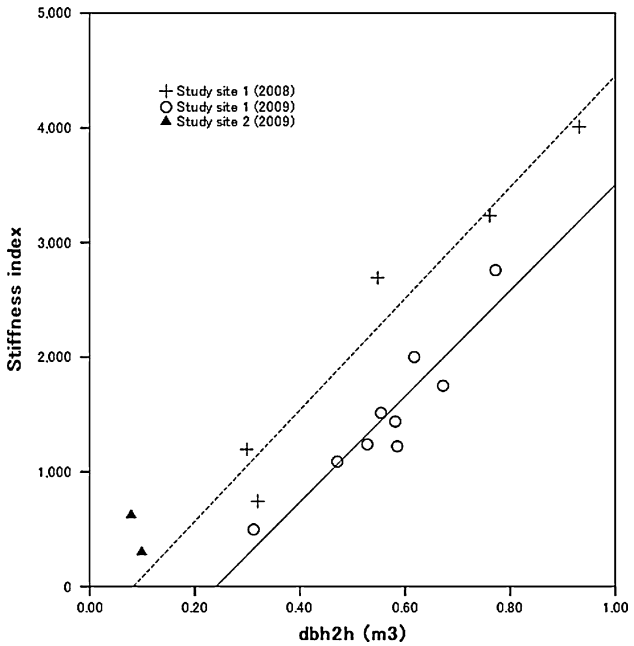


Fig. 3 Relationship between stiffness index (SI) and dbh^2h (m^3) for the uprooted trees from the tree-pulling experiments in 2008 and 2009. The *solid line* is a regression line for 2009 data on study site 1 ($y = 4611.9x - 1108.0$, $R^2 = 0.871$), and the *dotted line* is for 2008 data on study site 1 ($y = 4858.1x - 402.8$, $R^2 = 0.948$)

after water supply (14.2 ± 5.9 and 13.3 ± 7.0 kgf/cm^2) was not significant ($P > 0.05$). These results indicate that water supply reduced, in general, compressive strength in both the surface and deeper soil, excluding the deeper soil of study site 2.

On study site 1, d*bh*, dbh^2h, tree height, root plate area, and volume were significantly correlated with TM_{max} (Table 3, $P < 0.05$). No significant differences were found between $TM_{gravity}$ and other factors except stem angle at the maximum force. On study site 2, limited data did not make possible corresponding analyses.

Water contents inside and below the root plate were significantly correlated with TM_{max} on study site 1. Water content below the root plate was also associated with several factors important for root anchorage including $TM_{applied}$, β, and *SI*. This indicates that water content below the root plate strongly affected root anchorage under the combined effects from strong wind and high water content in the soil. In addition, since TM_{max} and water content below the root plate were negatively correlated, a tree was more easily uprooted with increasing water content (Fig. 5). On the other hand, due to the positive correlation between water content inside the root plate and TM_{max}, water content inside the plate prevented the root plate from being easily uprooted. In addition, increasing the water content inside the plate significantly decreased the normalized area of the hinge side (Fig. 6).

Discussion and conclusions

Various experimental procedures such as the quantity of supplied water and time waited after it before pulling down a tree were used to analyze how rapid rainfall in addition to strong wind would influence root anchorage. Our results indicated that the quantity of water did not directly relate to factors affecting root failure such as soil compressive strength and TM_{max}, although there were few exceptions. On the other hand, water supply effects were also found for the root failure mechanism.

The soil compressive strength was reduced due to the water supplied except for in the deeper soil of study site 2, although the extent of reduction was not directly related to the quantity of the supplied water, the time waited before pulling tree over, or the soil depth (Fig. 4). More importantly, water supplied to the soil clearly reduced *SI* (Fig. 3), which indicates that the key factors to reduce *SI* are water contents inside and below the root plate. Koizumi (1987) found that *SI* on wet soil was 20–30% lower than that observed for soil conditions without rainfall. Moreover, in this work, water content inside the root plate appeared to enhance root anchorage (Table 3).

During tree pulling, root weight is the second most important component for anchorage, because it prevents uprooting of the root plate (Coutts 1986). We also found that water content was more likely to influence $TM_{applied}$, which initially occurs due to the horizontal force from wind (see e.g., Nicoll et al. 2006). Therefore, water content inside the root plate contributes to the stability of root anchorage at the beginning of the failure process, likely due to the additional weight from supplied water. However, water content inside the plate also reduced plate area on the hinge side (Fig. 6). $TM_{gravity}$ was subsequently generated, because the center of gravity moved to the outside of the root plate (see e.g., Coutts 1986). The smaller the plate area of the hinge side, the higher $TM_{gravity}$ exerted at the maximum force. Thus, the reduced hinge-side plate area makes a tree unstable and probably leads to uprooting, although it is strongly dependent on root depth, soil characteristics, and water components in the soil.

In addition, TM_{max} was also negatively correlated with soil water content below the root plate (Fig. 5). These results suggest that water impacts the behavior of the root plate. One possible reason is that high water content produces high pore water pressure in the soil. Thus, water content below and underneath the root plate leads to hydraulic fracturing in the soil due to tree loading. The fracturation is subjected to a lower moment than the maximum turning moment (Rodgers et al. 1995). In addition, a crack first forms under the stem and spreads to windward and leeward positions (Coutts 1986). We assumed that this crack might be filled with water, leading to faster uprooting

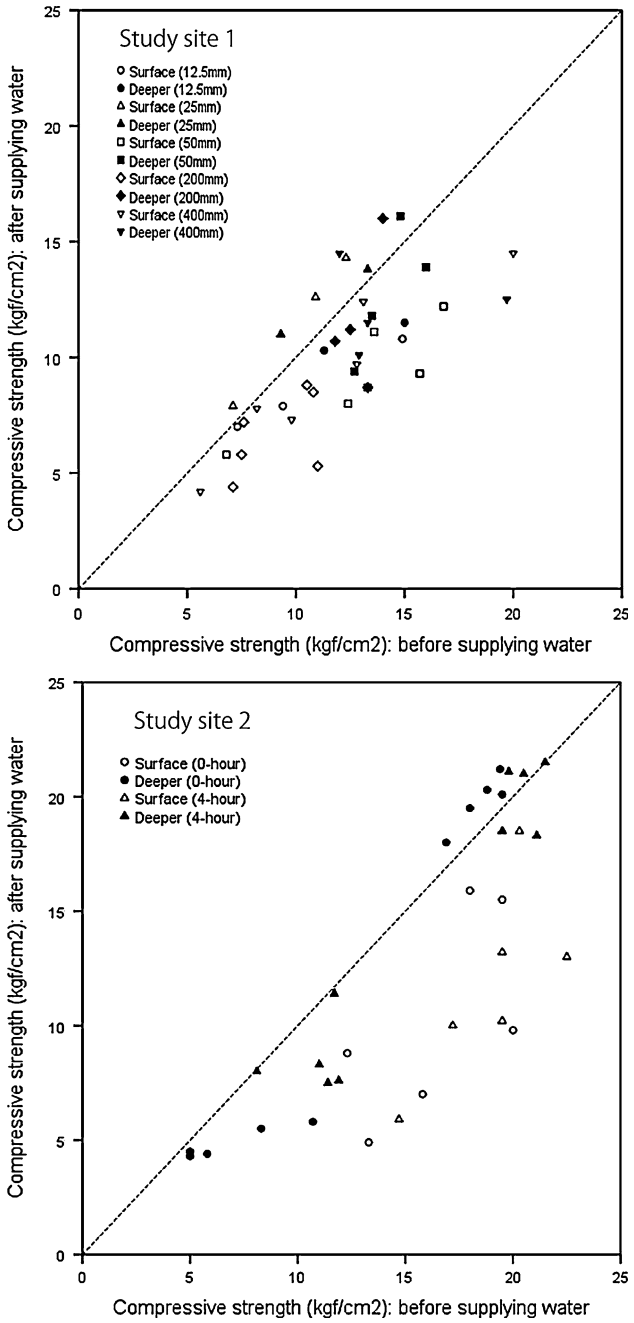


Fig. 4 Mean soil compressive strength (kgf/cm^2) before supplying and after supplying water depending on the soil depth for the soil surface (10–30 cm) and deeper soil (>30 cm)

of the root plate. Ray and Nicoll (1998) pointed out that water can enter the crack quickly, and then the saturated area is liquefied due to swaying caused by wind pressure.

Although our tree-pulling experiments did not account for oscillation from stem swaying caused by wind pressure, the relationship between water content below the root plate and TM_{max} is an important factor for root anchorage. Water content below the root plate is also strongly dependent not only on the quantity of supplied water but also on soil type and environmental conditions at any particular slope. More detailed studies, including water saturation of the root plate based on soil and rooting types, are required to clarify the critical quantity of rainfall enhancing tree failure as caused by strong wind.

Moreover, water content and tree behavior might be associated with failure type (i.e., uprooting and stem breakage). All trees were uprooted despite various quantities of water supplied on study site 1, while only two of five trees were uprooted and others were broken on study site 2. One of the broken trees was a control, and the other two trees received 100 mm of water with a 4-h waiting time for pulling. The results on study site 1 appeared similar to the previous findings of Koizumi (1987), who indicated a high likelihood of uprooting caused by strong wind with heavy rainfall based on tree pulling of five trees one night after a heavy rain. In our work, three trees were broken on study site 2, although the soil type (volcanic) and experimental situation (water supply plus waiting time) were similar to Koizumi's (1987) work. One possible reason for this difference may be that there was oscillation from stem to root caused by wind pressure, as tree-pulling experiments were conducted after a storm. Thus, additional work is required to understand in more detail the mechanism of tree failure including stem behavior during a typhoon and a storm. It is important to take into account the interaction among water content in the soil, rooting type and size, and oscillation from the sway of a stem.

Current wind damage models for forests are commonly using the relationship between TM_{max} and stem weight or dbh^2h (e.g., Achim et al. 2005; Cucchi et al. 2004; Nicoll et al. 2006; Peltola et al. 2000). However, a question arises whether these relationships are still suitable to analyze root anchorage under the combined loading of intensive rainfall and strong winds caused by a typhoon. Moreover, dbh^2h or stem mass represents only above-ground characteristics; hence, it does not fully account for below-ground characteristics, i.e., root anchorage (Achim and Nicoll 2009). Rapidly supplied water might also have a crucial effect on the relationships involved. For further studies, additional factors such as rainfall in a short period of time need to be carefully studied to understand in more detail the mechanism of damage caused by a typhoon and a storm.

Further investigation is also required to analyze how rapid rainfall produces additional factors influencing root anchorage. On study site 1, uprooting was the only failure type when water was supplied to the soil, whereas stem breakage was also observed on study site 2. The broken trees experienced a 4-h waiting period before pulling. For future experiments, soil structure and water profile throughout the soil and root plate should also be carefully observed. In particular, the quantity of precipitation might create various effects on root stability because of the

Table 3 Pearson's correlation coefficients (r) between factors related to root anchorage for 2009 data in study site 1

Factor 1	Factor 2	n	r	Sig.
[Study site 1 (Kamiatago)]				
TM_{max}	dbh	9	0.710	0.032
	dbh^2h	9	0.670	0.049
	Tree weight	9	0.683	0.043
	Root plate area	9	0.789	0.011
	Root plate volume	9	0.691	0.039
	Water content below the root plate	9	−0.728	0.026
	Water content inside the root plate	8	0.827	0.011
$TM_{applied}$	Root plate area	9	0.794	0.011
	Water content inside the root plate	8	0.896	0.003
	Water content below the root plate	9	−0.802	0.009
Stem angle at max force (β)	Water content below the root plate	9	0.712	0.032
Stiffness index (SI)	dbh^2h	9	0.934	0
	Water content below the root plate	9	−0.671	0.045

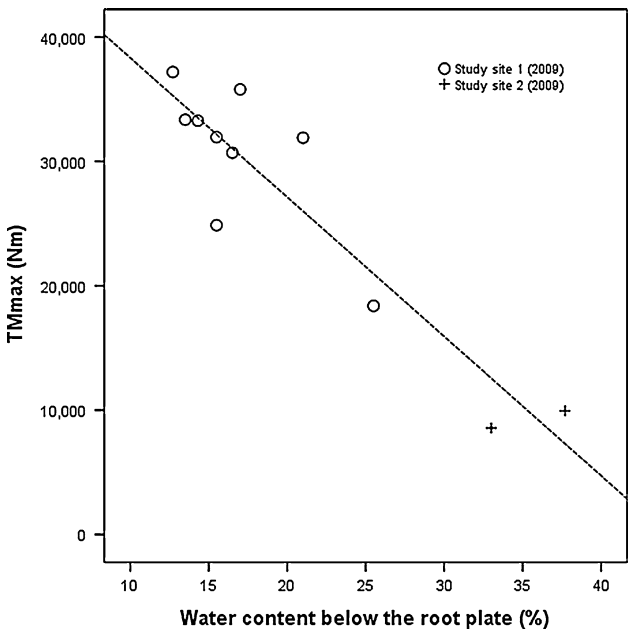


Fig. 5 Relationship of TM_{max} (Nm) and the water content below the root plate (%) for all target trees on study site 1 and the uprooted trees on study site 2. The line is based on the relationship found on study site 1 ($y = -1046.89x + 48448.09$, $R^2 = 0.530$)

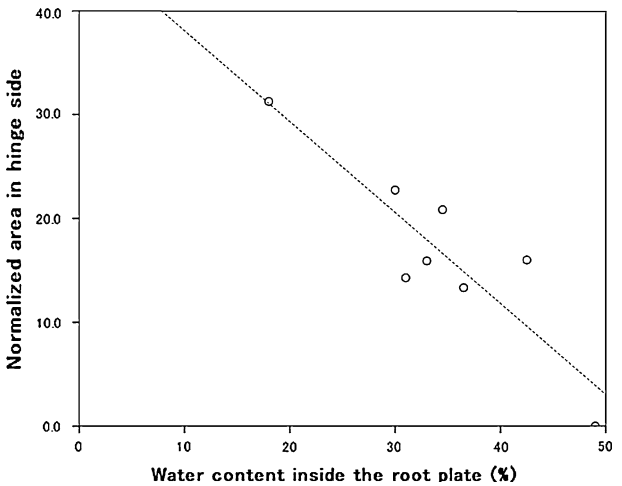


Fig. 6 Relationship between the normalized area of hinge (leeward) side against the root plate and the water content inside the root plate (%). The line is based on the relationship found on study site 1 ($y = -0.88x + 46.82$, $R^2 = 0.797$)

complexity of water movement in the soil, the heterogeneity of the soil structure, and effects from varying slopes. Stem swaying should also be taken into account with the water content, because oscillation of the root plate might accelerate liquefaction at the crack under the root plate due to it being filled with water.

To conclude, rapid rainfall on the soil in addition to strong wind pressure on a stem would influence the root anchorage mechanism. Since stiffness decreased as a result of additional water supplied to the soil, rainfall could be expected to influence the mechanism of root failure in addition to strong wind pressure. We found that water inside the root plate contributed to root anchorage during the early process of failure. Thus, it could be assumed that the total weight of the root plate increased with added water. This weight influenced $TM_{applied}$, which was due to horizontal force by pulling down. On the other hand, water content inside the root plate also reduced the root plate area of the hinge (leeward) side.

Trees would be more likely to uproot because the center of gravity is outside of the plate immediately after experiencing horizontal force. In addition, we found that high water content below the root plate decreased TM_{max}. Due to the significant relationship between $TM_{applied}$ and the water content below the root plate, we assume that the root

plate might slip on the crack created under the plate, which was filled with water. Therefore, the combined effects of rapid rainfall and strong wind might alter the process of failure and the root anchorage components compared with root anchorage as a result of wind pressure only.

For future studies, several factors relating to root anchorage should be investigated in more detail, such as for instance, soil structure, water profile throughout the soil, and root plate. Stem swaying should also be taken into account with the water content to understand the effects of oscillation of the root plate in more detail. Consideration of rapid rainfall is also needed to estimate long-term tree failure risk after a storm characterized by heavy rain.

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